

There is a compressed KNOPPIX file in *image/KNOPPIX*, which you need to uncompress, and then extract the required files to customise as follows:

```
$ extract_compressed_Is image/KNOPPIX/KNOPPIX > /tmp/knoppix_extracted
$ sudo mount -o loop /tmp/knoppix_extracted image
$ sudo cp -Rj image/* source/KNOPPIX
$ sudo umount image
$ sudo losetup -d /dev/loop0
```

Change boot image

Make a 640x400-pixel, 16-colour graphic and save it as a GIF (for example, *logo.16.gif*) using the GIMP. Feel free to create anything you like. All you have to take note of is to convert the image to a 16-colour graphic. Then we need to convert this GIF image to *lss16*. For this you require the *ppmtolss16* tool, which is part of the *syslinux* package in Debian:

```
$ gimpopen <logo.16.gif >logo.ppm
$ ppmtolss16 <logo.ppm > logo.16
```

Move the image *logo.16* to the master boot folder:

```
$ sudo mv logo.16 master/boot/isolinux/logo.16
```

Additionally, you can also take a look at the *master/boot/isolinux/boot.msg* file to change some text, if required.

Change the start-up script text

The start-up script text is the one that you see immediately after the boot screen. We can customise this text in order to suit your needs. This is how to access the script:

```
$ sudo gunzip master/boot/isolinux/minirt24.gz
$ sudo mount master/boot/isolinux/minirt24 image -t ext2 -o loop
```

Open the *image/linuxrc* file in a text editor and edit as required. Save it! Finally, repack it in as follows:

```
$ sudo umount image
$ sudo gzip -q master/boot/isolinux/minirt24
```

Modify the GRUB menu entry

Edit the file *source/KNOPPIX/boot/grub/menu.lst* to look like the following code:

```
default 0
timeout 2
title Stylezen OS
kernel /boot/linux24 root=/dev/hda1 quiet vga=normal noacpi nosmp nodma
noscsi frugal
initrd /boot/minirt24.gz
```

Now you have all the files required for customising DSL in your *dsl-sandbox* folder. We *chroot* into the source directory and do the required customisation. We have *vi*

to do all the editing tasks inside the *chroot* environment.

```
$ sudo chroot source/KNOPPIX/
bash-2.05b# mount -t proc /proc proc
bash-2.05b# rm etc/skel/.dmdesk/.MyDSL etc/skel/.dmdesk/Apps etc/skel/.dmdesk/.links/home dir
```

Desktop and browser changes

First, you need to remove all menu items from the default desktop—that is, JWM. In order to do this, edit the file named *opt/mydsl_menu/jwm/menu_template* and */etc/skel/jwmrc* inside the *chroot* environment so that its contents resemble the following code:

```
<?xml version="1.0"?>
<JWM>
</JWM>
```

One of the requirements is to start the OS and show Firefox as the default browser displaying some website. In order to achieve this, open the file */etc/skel/.xinitrc* and search for the word *dillo*. When you are at this line, edit it to resemble the following code:

```
firefox &&>/dev/null &
```

We may opt to remove the next line *torismo 2>/dev/null &* if we don't want to see system statistics on the desktop, by default.

Open the file */usr/local/firefox/defaults/pref/firefox-branding.js* and edit it to point to whatever URL you want, in order to provide a default homepage when Firefox starts up. Here is what my file looks like:

```
pref('startup.homepage_override_url','http://www.stylezen.org/');
pref('startup.homepage_welcome_url','http://www.stylezen.org/');
```

Have a look at */opt/backgrounds/readme.txt* in order to change the default desktop backgrounds.

After going through all the above steps, do not forget to unmount the */proc* filesystem and log out of the *chroot* environment, which is illustrated below:

```
bash-2.05b# cd /
bash-2.05b# umount proc
bash-2.05b# logout
```

Build the OS

Now that we're done with all the customisations, let's start building the OS. It's very simple. If you have followed the folder structure that I have mentioned in the beginning, then issue the following commands from the *dsl-sandbox/* directory:

```
$ sudo rm -rf source/KNOPPIX/rt_moved
$ sudo mkisofs -R -U -V "KNOPPIX.net filesystem" -publisher \
```